

The Level-2 Harnack Reduction and Sphere Absorption: localizing all N_2 negativity to two ball-absorbed coefficients

(Seed 4, Round 2, Layer 3)

Loop Experiment

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Abstract

We reduce the Numerator Positivity target $N_2 := (1 + q^2 + q^4)Q_2 \geq 0$ (rank 3, $\gcd(d, 3) = 1$) to a discrete Harnack inequality for the level-1 polynomial Q_1 on profile space, and show that *all* negativity of the Harnack form is concentrated at the two exponents 2 and 4, where it is absorbed *exactly* by EMD-sphere counts. In particular $[q^2]N_2(c) = 0$ identically. Everything becomes lattice point counting for a polygonal quasi-norm on the profile triangle. We prove a Cap-Compression Lemma making each fixed coefficient of the Harnack form a finite problem, and deduce $[q^j]N_2 \geq 0$ for all $j \leq 11$ and all d (modulo a sharp cap constant verified for $d \leq 35$; unconditionally for $j \leq 5$). The same structure is verified at level 3, yielding a uniform Sphere Absorption Conjecture for all levels n .

1 Setup and inherited results

Standing notation of the Layer-2 synthesis: profiles $c = (c_0, c_1, c_2)$, $c_i \geq 0$, $\sum c_i = d$, $\gcd(d, 3) = 1$; $\text{EMD}(c, c') = 3 \max(0, c'_1 - c_1, c_0 - c'_0) + (c'_0 - c_0) - (c'_1 - c_1)$; $H_{c,m} = (q; q)_m F_{c,m}$. We use as GREEN inputs: the H -recursion $(1 + q^m + q^{2m})H_{c,m} = \sum_{c'} q^{m \text{EMD}(c', c)} H_{c', m-1}$ with $H_{c,0} = 1$ (Layer 2, Seeds 2/3/4); the Q -transform $Q_n = \sum_{m \leq n} (-1)^{n-m} q^{\binom{n-m}{2}} \begin{bmatrix} n \\ m \end{bmatrix} H_{c,m}$ (Seed 3); and $Q_1 \geq 0$ (Layer 1).

Throughout, $e' := \text{EMD}(c', c)$ (distance *into* c), and $b_e(c) := \#\{c' : \text{EMD}(c', c) = e\}$ are the *sphere counts*; $n_1 := b_1$, $n_2 := b_2$. The *ball gen-*

erating function is $B_c(q) := \sum_{c'} q^{\text{EMD}(c',c)} = (1 + q + q^2) H_{c,1}$ (the $m = 1$ recursion), so $H_1(c) = B_c(q)/(1 + q + q^2)$ and $Q_1(c) = H_1(c) - 1$.

2 The Harnack reduction

Theorem 1 (Identities T1/T2). *For every profile c ,*

$$N_2(c) = \sum_{c' \neq c} q^{2e'} H_1(c') + (q + q^3 + q^5) - (q + q^2 + q^3 + q^4 + q^5) H_1(c), \quad (\text{T1})$$

$$N_2(c) = \underbrace{[B_c(q^2) - 1 - q^2 - q^4]}_{\text{ball term}} + \underbrace{\left[\sum_{c' \neq c} q^{2e'} Q_1(c') - q(1 + q + q^2 + q^3 + q^4) Q_1(c) \right]}_{=: \text{har}(c)}. \quad (\text{T2})$$

Proof. The Q -transform at $n = 2$ reads $Q_2 = H_2 - (1 + q)H_1 + q$. Multiply by $1 + q^2 + q^4$ and substitute the $m = 2$ recursion $(1 + q^2 + q^4)H_2(c) = \sum_{c'} q^{2\text{EMD}(c',c)} H_1(c')$:

$$N_2(c) = \sum_{c'} q^{2e'} H_1(c') - (1 + q)(1 + q^2 + q^4)H_1(c) + q(1 + q^2 + q^4).$$

Since $(1 + q)(1 + q^2 + q^4) = 1 + q + q^2 + q^3 + q^4 + q^5$ and the $c' = c$ term of the sum is $H_1(c)$ (as $\text{EMD}(c, c) = 0$), the constant-1 part cancels, giving (T1). Substituting $H_1 = 1 + Q_1$ into (T1): the “1” parts contribute $\sum_{c' \neq c} q^{2e'} = B_c(q^2) - 1$ from the sum and $-(q + \dots + q^5) + (q + q^3 + q^5) = -q^2 - q^4$ from the remaining terms, giving (T2). Both identities were verified exactly for $d = 4, 5, 7$, all profiles (`seed4_R2L3_engine.py`). $\square$

3 Sphere geometry

In deviation coordinates $s = c'_0 - c_0$, $t = c'_1 - c_1$, the distance into c is the polygonal quasi-norm $f(s, t) = 3 \max(0, -t, s) - s + t$; the profile constraints are $s \geq -c_0$, $t \geq -c_1$, $s + t \leq c_2$. By the mod-3 EMD lemma (Layer 2, GREEN), $f(s, t) \equiv t - s \pmod{3}$: the residue of f is determined by the position class $\Lambda_j = \{t - s \equiv j \pmod{3}\}$.

Proposition 1 (Spheres of radius 1 and 2). *(i) $\{f = 1\} = \{(0, 1), (-1, 0), (1, -1)\}$, valid in the triangle iff $c_2 \geq 1$, $c_0 \geq 1$, $c_1 \geq 1$ respectively. Hence $n_1(c) =$*

$\text{rank}(c) := \#\{i : c_i \geq 1\}$. (ii) $\{f = 2\} = \{(0, 2), (-1, 1), (-2, 0), (1, 0), (0, -1), (2, -2)\}$ with validity thresholds $c_2 \geq 2, c_0 \geq 1, c_0 \geq 2, c_2 \geq 1, c_1 \geq 1, c_1 \geq 2$; a short case analysis on (c_2, c_0, c_1) gives $n_2(c) \geq 2$ for all $d \geq 2$.

Proof. (i) Solve $f(s, t) = 1$ on the three linearity sectors: sector $s \leq 0, t \geq 0$: $t - s = 1$ gives $(0, 1), (-1, 0)$; sector $s \geq \max(0, -t)$: $2s + t = 1$ gives $(1, -1)$ (and $(0, 1)$ again); sector $t \leq 0, s \leq -t$: $-2t - s = 1$ gives no new points. Validity: $(0, 1)$ means $c' = c + (0, 1, -1)$, needing $c_2 \geq 1$; similarly for the others. (ii) Same enumeration at $f = 2$. For the count: if $c_2 \geq 2$ then $(0, 2), (1, 0)$ are valid; if $c_2 = 1$ then $(1, 0)$ plus one of $(-1, 1), (0, -1)$ (as $c_0 + c_1 \geq 1$); if $c_2 = 0$ then $d \geq 2$ forces two valid points among $(-1, 1), (-2, 0), (0, -1), (2, -2)$ in each of the cases $c_0 \geq 2; c_0 = 1, c_1 \geq 1; c_0 = 0, c_1 \geq 2$. $\square$

Proposition 2 (Low coefficients of Q_1). *Writing $A_j(c) := \#\{c' : f \leq j, f \equiv j \pmod{3}\}$, we have $H_1 = B_c \frac{1-q}{1-q^3}$, hence $[q^j]H_1 = A_j - A_{j-1}$ and in particular*

$$[q^0]Q_1 = 0, \quad [q^1]Q_1 = n_1 - 1, \quad [q^2]Q_1 = b_2 - b_1, \quad [q^3]Q_1 = 1 + b_3 - b_2.$$

Proof. $B_c/(1-q^3)$ has j -th coefficient $\sum_{k \geq 0} b_{j-3k} = \#\{f \leq j, f \equiv j \pmod{3}\} = A_j$ because f -values in a fixed residue class below j are exactly $j, j-3, j-6, \dots$; multiplying by $1-q$ gives $A_j - A_{j-1}$. Then $A_0 = b_0 = 1, A_1 = b_1, A_2 = b_2, A_3 = b_0 + b_3$. $\square$

4 Exact absorption at q^2 and q^4

Theorem 2 (Vanishing at q^2). $\text{har}_2(c) = -(n_1(c) - 1)$ *identically; since the ball term has $[q^2] = n_1 - 1$, we get $[q^2]N_2(c) = 0$ for every profile and every d .*

Proof. $[q^2] \sum_{c' \neq c} q^{2e'} Q_1(c') = \sum_{e'=1} [q^0] Q_1(c') = 0$, and $[q^2] q(1 + \dots + q^4) Q_1(c) = [q^1] Q_1 + [q^0] Q_1 = n_1 - 1$ by Proposition ?? $\square$

Theorem 3 (Absorption at q^4). $\text{har}_4(c) = \sum_{c': e'=1} (b_2(c') - b_1(c')) - b_3(c)$, and $\text{har}_4(c) \geq -(n_2(c) - 1)$; hence $[q^4]N_2(c) \geq 0$.

Proof. The formula: $[q^4] \sum_{c' \neq c} q^{2e'} Q_1(c') = \sum_{e'=1} [q^2] Q_1(c') + \sum_{e'=2} [q^0] Q_1(c') = \sum_{e'=1} (b_2 - b_1)(c')$, while $[q^4] q(1 + \dots + q^4) Q_1(c) = \sum_{i=1}^3 [q^i] Q_1(c) = b_3(c)$ (telescoping, Proposition ??). All quantities depend on c only through $(\min(c_i, 3))_i$ (all validity thresholds are ≤ 3 ; neighbours shift coordinates

by ≤ 1 , absorbed in the cap), so the inequality for all d follows from its verification on every capped class, which is contained in the exact sweep $d \leq 35$ (`seed4_R2L3_bigsweep.py`: no failures). $\square$

5 Cap-Compression and the low band

Lemma 1 (Cap-Compression). *For every fixed j , $\text{har}_j(c)$ depends on c only through the capped triple $(\min(c_i, M_j))_i$ with $M_j = 2j$. Empirically the sharp value is $M_j = j$ (verified for $j \leq 12$, all $d \leq 35$, $3 \nmid d$).*

Proof sketch. har_j involves $[q^{j-2e'}]Q_1(c')$ for $2e' \leq j$ and $[q^{j-i}]Q_1(c)$, $i \leq 5$. Since $f \geq \max(|s|, |t|, |s+t|)/1$ on each sector, profiles with $e' \leq j/2$ have coordinates within j of c 's, and $[q^k]Q_1(c') = A_k - A_{k-1}$ counts points within f -distance $k \leq j$ of c' , whose triangle-validity thresholds are $\min(c'_i, k)$. All thresholds appearing are therefore at most $2j$ in terms of c 's coordinates. $\square$

Theorem 4 (Low band). *For all d with $\gcd(d, 3) = 1$ and all profiles c : $[q^j]N_2(c) \geq 0$ for $j \leq 5$ unconditionally, and for $j \leq 11$ modulo the sharp cap constant $M_j = j$ of Lemma ???. Moreover $\text{har}_j(c) \geq 0$ for $j \leq 11$, $j \notin \{2, 4\}$ (same caveat).*

Proof. By Lemma ?? each statement at fixed j is a function of the capped class; every capped class with caps $\leq M$ is realized by a profile with $d \leq 3M + 1$, $\gcd(d, 3) = 1$. The exact sweep over all $d \leq 35$, $3 \nmid d$ (24 values, all profiles, exact $\mathbb{Z}[q]$ arithmetic) found no violation. For $M_j = 2j$ this covers $j \leq 5$; for the sharp $M_j = j$ it covers $j \leq 11$. $\square$

6 The remaining wall and the level- n lift

Conjecture 1 (S1/S2, level 2). *For all c and $j \notin \{2, 4\}$: $\text{har}_j(c) \geq 0$. (Together with Theorems ??-?? and the ball term this implies $N_2 \geq 0$.) Verified exactly for all $d \leq 35$ with $3 \nmid d$, all profiles, all j .*

Tightness data: $\text{har}_j = 0$ occurs at $j \in \{0, 1, 3, 5, 6\}$ and in a wide top band (e.g. odd $j \in [27, 39]$ at $d = 7$); no coarse bound can work there. Reciprocity fails: $q^{\deg} X(1/q)$ for $X \in \{Q_1, \text{har}, N_2\}$ matches no profile transform, so the top band is not a dual low band. The involution substrate: each unit move $(0, 1), (-1, 0), (1, -1)$ on the triangle shifts the position class $t - s \bmod 3$ by $+1$ while changing f by $+1$ or -2 (the Preimage EMD Dichotomy in geometric form).

Conjecture 2 (Sphere Absorption, all levels). *In the level- n analogue of (T2) (proved as an identity at $n = 3$: $N_3(c) = \sum_{c' \neq c} q^{3e'} H_2(c') + q(1 + q + q^2)(1 + q^3 + q^6)H_1(c) - (q + \dots + q^8)H_2(c) - (q^3 + q^6 + q^9)$, then split off constants), the Harnack form $\text{har}^{(n)}(c)$ has negative coefficients only at the exponents $n \cdot e$ for small e , each bounded below by $-(b_e(c) - 1)$ and absorbed exactly by the ball term $B_c(q^n) - 1 - q^n - \dots$; moreover $\text{har}_n^{(n)} = -(n_1(c) - 1)$ exactly, so $[q^n]N_n(c) = 0$. Verified at $n = 2$ ($d \leq 35$) and $n = 3$ ($d = 4, 5$).*

Status

GREEN: Theorems ??, ??, Propositions ??, ??; Theorem ?? and the $j \leq 5$ part of Theorem ?? are GREEN modulo writing out the finite cap case lists (machine-checked exhaustively). YELLOW: sharp cap $M_j = j$; Conjectures ??, ??. OPEN WALL: $\text{har}_j \geq 0$ for unbounded j (middle/top band) — a lattice point-counting inequality for the polygonal quasi-norm f on the triangle, quasi-polynomial in (j, c) , hence in principle decidable by finite region-by-region verification (Ehrhart route).